

# Can't touch this? Why vibrotactile feedback matters in educational VR

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## ABSTRACT

This study investigates the relationship between vibrotactile feedback and sense of presence in VR. The inquiry focuses on corrective and reinforcing feedback in STEM learning outcomes using a VR environment called Looking Inside Cells. In a randomized within-subject design experiment (N=68) participants got assigned to a vibrotactile and non-vibrotactile condition. Our hypotheses: Participants in the vibrotactile-condition report higher sense of presence ratings compared to the non-haptic condition. Results indicate that vibrotactile feedback increases the sense of presence and impacts metacognition. Participants who received corrective feedback as a vibrotactile stimuli are more likely to underestimate their actual test performance.

**Keywords:** Virtual Reality, Vibrotactile Feedback, Metacognition, Human Computer Interaction, Haptics

## 1 INTRODUCTION

This research study investigated if vibrotactile feedback in VR represent a distinct and powerful form of stimuli that enhances the subjective experience and engages our senses. The key argument of this paper is that vibrotactile stimuli used as educational feedback do not only bear the potential to increase sense of presence but also learning and self-efficacy.

### 1.1 Research Questions

This research investigated two hypotheses with a randomized within-subject design:

RQ1: Learners with a high sense of presence achieve higher scores on the assessment.

RQ2: Learners in the vibrotactile condition report a higher sense of presence compared to the non-haptic condition.

## 2 THEORY

This study draws from two theoretical frameworks, the multimedia learning principles and the dual coding theory. The dual coding theory [1] states that different types of stimuli impact learning. A non-verbal channel and a verbal channel are available to process information. The related cognitive load theory [2] proposes that channels have limited processing capacity.

Mayer's multimedia learning principles [3] provide guidelines on how to design multimedia learning experiences.

### 2.1 Immersion & Presence

Immersion and presence are two constructs, which are often mentioned in the context of VR [4]. Immersion refers to the objective dimension of technological features like refresh rate and

resolution. Presence as a subjective quality relates to the technology's ability to evoke affection in the user. The smaller the gap in latency, resolution and spatial stimuli, the higher is the immersion.

### 2.2 Feedback

Formative feedback is directive and facilitative. It provides clear indicators if a specific goal was achieved [5].

Scholars proposed that "negative reinforcement", informing learners about wrong actions, is more effective than rewarding learners ("positive reinforcement") [6]. This contradicts Skinner's findings and his concept of operant conditioning [7].

Does haptic feedback impact presence or improve learning outcomes? The reviewed literature offers mixed results. [8].

### 2.3 Metacognition and self-assessment

Metacognition, often referred to as "thinking about thinking," is a cognitive process that plays a pivotal role in education [9].

One of the key aspects of metacognition is self-assessment. It involves students evaluating their own learning progress, knowledge, and skills. Metacognition and self-assessment are closely intertwined. When students engage in self-assessment, they are essentially applying metacognitive processes to evaluate their own learning [10].

## 3 METHOD

### 3.1 Participants and Experimental Design

Participants (n=68) were recruited via flyers around New York University's campus (35 women, ages: 17 – 39, M = 25 SD = 4). 43 identified as Asian, 13 as White, 5 as Black or African American, 5 as South Asian and 2 as hispanic.

34 participants were randomly assigned to the corrective feedback condition and 34 were assigned to the reinforcing feedback condition. Both groups finished a tutorial, a build level and a mitosis level in VR (Table 1).

Table 1. Experimental design

| Level/<br>Condition | Tutorial     | Build level<br>(corrective<br>feedback) | Mitosis level<br>(reinforcing<br>feedback) |
|---------------------|--------------|-----------------------------------------|--------------------------------------------|
| Corrective          | Vibration On | Vibration On                            | Vibration Off                              |
| Reinforcing         | Vibration On | Vibration Off                           | Vibration On                               |

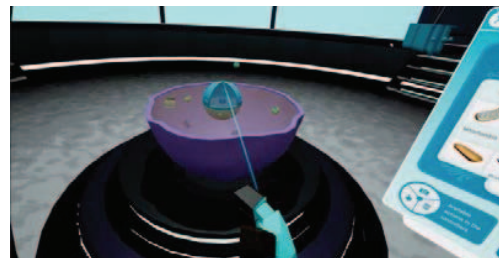


Figure 1: Screenshot of the Build a cell level

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Both groups experienced the same Tutorial level. The corrective feedback group received vibrations in the Build level but no vibrations in the Mitosis level. The reinforcing group did not receive vibrations in Build but in the Mitosis level.

#### 4 RESULTS

The table below summarizes the mean and standard deviations of the variables of interest.

Table 2. Mean and SD of the variables of interest

| Group/<br>Variable          | Corrective<br>Group | Reenforcing<br>Group |
|-----------------------------|---------------------|----------------------|
| Sense of presence (0 to 90) |                     |                      |
| Build level                 | 71.6 (11)*          | 68.3 (12.2)*         |
| Mitosis level               | 67.6 (11.4)*        | 70.4 (12.9)*         |
| Total score (out of 14)     | 8.3 (2.6)           | 8.7 (2.4)            |
| Self-assessment             | 2.8 (2.6)**         | 1.7 (2)**            |

\* within-group difference \*\* between-group difference

##### 4.1 Group comparisons

A within-group comparison showed that the corrective group rated the vibrating Build level higher than the non-vibrating Mitosis level ( $d=0.3$ ) (Fig. 2). The reinforcing group rated the vibrating Mitosis level higher than the non-vibrating Build level, ( $d=0.3$ ). Participants tend to attribute higher sense of presence ratings to the level with enabled vibrotactile feedback. The difference between the corrective group's presence ratings approached statistical significance,  $t(33) = 1.82$ ,  $p = .077$ . The difference between reinforcing group's presence ratings was statistically significant,  $t(33) = 2.03$ ,  $p = .05$ .

Both groups performed almost the same on the post-assessment.

Comparing the groups' self-assessment revealed a significant difference,  $t(66) = 1.96$ ,  $p = .05$ . On average the corrective group underestimated their assessment score by 2.8 points.

##### 4.2 ANCOVAs

We conducted two analyses of covariances (ANCOVA), one for the Build level ( $R^2=.47$ ), which used corrective feedback and one for the Mitosis level ( $R^2=.25$ ), which relied on reinforcing feedback. Build ANCOVA

The first ANCOVA was conducted with group (corrective, reinforcing) as independent variable, pre-test, involvement and self-assessment as a covariate, and the post-assessment score of the Build level as the dependent variable.

The result revealed no effect of condition on the post score but an effect of sense of presence, involvement and self-assessment after controlling for prior knowledge,  $F(5, 62) = 9.9$ ,  $p < .05$ .

The pre-assessment score explains most of the proportion of variance in the post-test Build scores ( $\eta^2p = .33$ ). The covariate involvement ( $\eta^2p = .08$ ), sense of presence ( $\eta^2p = .07$ ) and self-assessment ( $\eta^2p = .19$ ) had a significant effect on the Build score.

#### 5 DISCUSSION

##### 5.1 The paradox of corrective feedback as vibrotactile stimuli

Our research showed that vibrotactile stimuli tend to increase the participant's sense of presence. An increased sense of presence is a beneficial precondition for learning. Previous research with

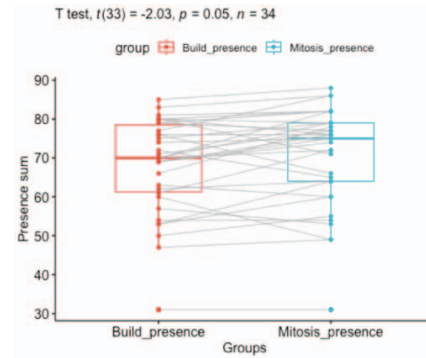


Figure 2: Reinforcing Group Presence Ratings (Boxplots)

traditional learning materials suggests that corrective feedback is superior to reinforcing feedback to increase learning outcomes. Haptic feedback in VR might pose a challenge and contradicts this traditional assumption that corrective feedback is more effective. We loaded vibrotactile stimuli with corrective feedback, in the Build level, and reinforcing feedback in the Mitosis level.

The interesting finding is that satisfaction and self-assessment correlated positively with a high sense of presence. Students who received corrective feedback in form of "punishing" vibrotactile stimuli are more likely to underestimate their performance on the post-assessment but report an overall higher satisfaction with their actual test score.

A between group comparison revealed no significant differences in assessment scores. The groups differed significantly in self-assessment and satisfaction. Those observations imply that vibrotactile feedback in our study had no impact on learning outcomes but on metacognitive processes, which in turn might affect learning in the long run.

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